

DEV_IMPL_GUIDE - FOUNDATION_FILE_STORAGE

1. Purpose

This guide explains how to implement the file/object storage foundation.

Use it with FOUNDATION_FILE_STORAGE.md.

This is a provisioning/runtime-setup guide. It is intentionally written as infrastructure setup steps rather than a business request execution flow.

2. What You Are Building

You are deploying object storage for SOPHIX:

- MinIO or AWS S3.
- Buckets.
- Object key conventions.
- Access policies.
- Retention/lifecycle rules.
- Upload/download patterns.

3. Step 1 - Choose Backend

Options:

Backend	When
MinIO	On-prem or self-managed object storage.
AWS S3	AWS deployment.

Application code should use an abstraction so the backend can change.

4. Step 2 - Create Buckets

Example buckets:

```
sophix-documents
sophix-invoices
sophix-exports
sophix-workorders
```

Rules:

- Do not store files in PostgreSQL.
- Store object keys in module DB tables.
- Keep buckets private.

5. Step 3 - Define Object Keys

Use predictable keys:

```
<operator>/<module>/<yyyy>/<mm>/<entity-id>/<file-id>.<ext>
```

Example:

```
WIK/billing/2026/05/inv_01HZ/file_01HZ.pdf
```

6. Step 4 - Implement Upload Flow

Recommended flow:

1. API validates auth.

2. API creates file metadata row.
3. API uploads object or returns pre-signed upload URL.
4. API stores object key, checksum, size, content type.
5. API returns file ID.

Metadata fields:

```
file_id
operator_code
module_code
object_key
content_type
size_bytes
checksum
created_at
created_by
```

7. Step 5 - Implement Download Flow

Recommended:

- API checks auth and operator scope.
- API returns stream or short-lived pre-signed URL.
- Never make bucket public.

8. Step 6 - Lifecycle And Retention

Configure:

- Retention by document type.
- Expiry for temporary exports.
- Versioning if required.
- Object lock only for documents that legally require immutability.

9. AWS Option

If using AWS:

- Use S3 private buckets.
- Enable encryption at rest.
- Use bucket policies and IAM roles.
- Enable lifecycle rules.
- Enable access logs where required.

10. Tests

Test:

- Upload works.
- Download requires authorization.
- Wrong operator cannot download.
- Metadata row matches object key/checksum.
- Deleted/expired object is handled clearly.

11. Common Mistakes

- Public buckets.
- Storing raw files in DB.

- Missing content type and checksum.
- Long-lived pre-signed URLs.
- Object keys without operator/module partitioning.